Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same. As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement

Ans) The business requirement is to predict insurance charges based on customer information such as age, sex, BMI, number of children, and smoking status.

- Since the dataset contains both the input and output features which are clearly defined, the **supervised machine learning** can be used to predict the insurance charges.
- The target variable, insurance charges, is a continuous numerical value, making it a **regression learning**.
- Categorical features like sex and smoker can be converted into numerical values using one hot encoding.

2.) Tell basic info about the dataset (Total number of rows, columns)

Ans)

- Total number of rows: 1339
- Total number of columns: 6
- Total independent variables 5
- Total dependent variables: 1

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Ans)

Feature	**Type**
Age	Numerical
Sex	Categorical
BMI	Numerical
Children	Numerical
Smoker	Categorical
Charges	Target

There are 2 input columns whose value has categorical/nominal data: Sex and Smoker

Since there are no ranking on these 2 categorical columns, need to use "One hot encoding" to convert the 2 columns from string to number using **get_dummies** method

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Ans) Random Forest ML algorithm with the following parameters provides the best R2_Score of 0.8970182967451767

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

#	Algorithm	Hyper Parameters	R2_Score
1	Multiple Linear Regression	default	0.7974504524086945
2	Support Vector Machine	Kernel=linear	-0.08705343840598312
3	Support Vector Machine	Kernel=rbf	-0.10353063408199814
4	Support Vector Machine	Kernel=poly	-0.0729269969753068
5	Support Vector Machine	Kernel=sigmoid	-0.1061359604346106
6	Support Vector Machine	Standardization	-0.012166509835097594
7	Support Vector Machine	Penalty Parameter (C=10)	-0.04248495679947917
8	Decision Tree	default	0.69756540429273
9	Decision Tree	criterion="squared_error", splitter="best", max_depth=2	0.8553674814329228
10	Decision Tree	criterion="absolute_error", splitter="random", max_depth=2	0.8000438773087624
11	Decision Tree	criterion="squared_error", splitter="random", max_depth=2	0.7917011754508682
12	Decision Tree	criterion="poisson", splitter="random", max_depth=2	0.7404533640713213
13	Random Forest	n_estimators=50	0.8666048997930276
14	Random Forest	n_estimators=100, criterion='squared_error', max_depth=10, random_state=0	0.8746551354890264
15	Random Forest	n_estimators=100, criterion='absolute_error', max_depth=10, random_state=0	0.887073000165453

16	Random Forest	n_estimators=100, criterion='poisson', max_depth=10, random_state=0	0.8773030794426256
17	Random Forest	n_estimators=100, criterion='poisson', max_depth=50, random_state=0	0.8725459695393039
18	Random Forest	n_estimators=100, criterion='poisson', max_depth=5, random_state=0	0.8955097801966453
19	Random Forest	n_estimators=100, criterion='squared_error', max_depth=5, max_features=5, random_state=0	0.8954436383936902
20	Random Forest	n_estimators=100, criterion='squared_error', max_depth=5, max_features=5, min_samples_split=5, random_state=0	0.8954882499619259
21	Random Forest	n_estimators=100, criterion='squared_error', max_depth=5, max_features=5, min_samples_split=5,bootstrap=True, random_state=0	0.8954882499619259
22	Random Forest	n_estimators=100, criterion='squared_error', max_depth=5, max_features=5, min_samples_split=5,bootstrap=True, random_state=10	**0.8970182967451767**
23	Random Forest	n_estimators=100, criterion='squared_error', max_depth=5, max_features=5, min_samples_split=5,bootstrap=True, random_state=42	0.8951552837653495

6.) Mention your final model, justify why you have chosen the same. Kindly create Repository in the name Regression Assignment.

Upload all the ipynb and final document in the pdf Communication is important (How you are representing the document.)

Ans) Justification:

Algorithm	**R^2 Score**	**Interpretation**
Multiple Linear Regression	0.797	Good baseline model but assumes a linear relationship.
SVM	Negative scores	Model failed to learn the relationship with the chosen settings.

Decision Tree	0.8553674814329228	Better because it captures non-linear relationships, but a single tree can overfit or underfit.
Random Forest	**0.8970182967451767**	Best because it combines many trees and captures complex relationships while reducing overfitting.

Github: 2.MachineLearning-Regression/Assignments at main · AnusuyaPonnusamy/2.MachineLearning-Regression